Status	Finished
Started	Saturday, 10 October 2026, 11:35 AM
Completed	Saturday, 10 October 2026, 11:38 AM
Duration	3 mins 36 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Bubble Sort
- b. Quick Sort
- c. Linear Search
- d. Merge Sort

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting is rarely used in programming
- b. Sorting decreases the efficiency of selection operations
- c. Sorting makes it harder to search for items
- d. Sorting helps in finding duplicates quickly

The correct answer is: Sorting helps in finding duplicates quickly

Question **3**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. Each part is sorted independently
- b. The parts are ignored
- c. Each part is searched for a specific element
- d. The parts are merged without sorting

The correct answer is: Each part is sorted independently

Question **4**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Merge Sort
- b. Bubble Sort
- c. Linear Search
- d. Binary Search

The correct answer is: Merge Sort

Question **5**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It recursively sorts subproblems
- b. It bubbles up the largest element in each pass
- c. It divides the list into sublists
- d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass

Question **6**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Insertion
- b. Distribution
- c. Extraction
- d. Selection

The correct answer is:
Insertion

Question **7**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Quick Sort
b. Bubble Sort
c. Merge Sort
d. Insertion Sort

The correct answer is: Bubble Sort

Question **8**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It allows for faster searching
b. It increases the number of comparisons needed
c. It has no effect on the search operation
d. It makes the search operation slower

The correct answer is: It allows for faster searching

Question **9**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. order()
b. arrange()
c. sorted()
d. sort()

The correct answer is: sort()

Question **10**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the largest element to its correct position
- b. Sorting the entire list in one pass
- c. Dividing the list into halves
- d. Bubbling up the smallest element

The correct answer is: Bubbling up the largest element to its correct position

Question **11**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sort()
- b. order()
- c. sorted()
- d. arrange()

The correct answer is: sorted()

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Merge Sort
- b. Quick Sort
- c. Insertion Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **13**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It requires external libraries
b. It is less efficient than custom sorting algorithms
c. It sorts data out of the box efficiently
d. It only works with integer arrays

The correct answer is: It sorts data out of the box efficiently

Question **14**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes it easier to select items based on their relationship to the rest of the items
b. It makes the data unsorted
c. It complicates the selection of items
d. It slows down the process

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **15**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Merge sort
b. Heap sort
c. Quick sort
d. Selection sort

The correct answer is:
Merge sort